Question no.1
Given the sequence, 150, 300, 450, 600, …, find the 15th term.

Solution: $a = \mathbf{150}$, $d = \mathbf{150}$, $n = 15$

So $a_{15} = 150 + 150(14) = 2250$

Question no.2
What is the general term of the sequence 1,8,27,64,125….

Solution:
$a_n = n^3$

Question no.3
The 3rd and the 4th terms of a geometric progression are 18 and 6 respectively. Find the sum to infinity of the progression.

Solution:

Given that $ar^2 = 18$ and $ar^3 = 6$

$r = 6/18 = 1/3$.

$a(1/9) = 18$

$a = 162$

$S_\infty = \frac{a}{1-r} = 243$

Question no.4

If first term $a = 4$, common difference $d = 2$, then what is the value of $10^{\mathbf{th}}$ term in an arithmetic sequence?
Solution:
a10= a + 9d = 4 + 9(2) = 22

Question no.5
Suppose $x + 2a,\ x + 4a,\ x + 6a, \ldots$, is an arithmetic sequence . Write its corresponding arithmetic series.

Hint:

Just sum (add) all the terms to find the corresponding series.

Question no.6

If $n = 1, 2, 3, 4, 5, 6, \ldots.$ then find the first four terms of the sequence $a_n = (n+2)^2$.

Solution:

$\because \quad n = 1, 2, 3, 4, 5, 6, \ldots.$

$\because \quad a_n = (n+2)^2$

$$\left.\begin{array}{ll} For\ n=1 \Rightarrow & a_1 = (1+2)^2 = (3)^2 = 9 \\ For\ n=2 \Rightarrow & a_2 = (2+2)^2 = (4)^2 = 16 \\ For\ n=3 \Rightarrow & a_3 = (3+2)^2 = (5)^2 = 25 \\ For\ n=4 \Rightarrow & a_4 = (4+2)^2 = (6)^2 = 36 \end{array}\right\}$$

Question no.7

Determine whether the sequence 9, 11, 13, 15, 17, ... is arithmetic or geometric. Give reason for your answer.

Solution:

First find the difference between each pair of consecutive terms.

11-9 = 2

13-11 = 2

15-13 = 2

17-15 = 2

Common difference = $d = 2$

The difference between the consecutive terms of the sequence is constant. The constant difference is called common difference. Hence the sequence is arithmetic.

Question no.8

Find the 5th term of the geometric sequence with $a_1 = 20$ and $a_2 = 40$.

- Solution:

  $r = 2^{nd}$ *term/ 1st term=40/20=2*

  $a_5 = 20 \bullet (2)^4 = 20 \bullet 16 = 320$

Question no 9.

$$\sum_{i=0}^{2} 2a_i$$

If $a_0 = 3,\ a_1 = 4,\ a_2 = -2,$ and $a_3 = 1$, then compute the summation:

$\because \quad a_0 = 3 \quad a_1 = 4,\ a_2 = -2,$ and $a_3 = 1$

$$\Rightarrow \sum_{i=0}^{2} 2a_i = 2\left(a_0 + a_1 + a_2\right)$$

$$= 2(3 + 4 - 2)$$
$$= 2(5)$$
$$= 10$$